



(12) **United States Patent**
Kitazume

(10) **Patent No.:** **US 12,388,408 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **FILTER AND MULTIPLEXER**

(56) **References Cited**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 253 days.

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(21) Appl. No.: **18/182,579**

(22) Filed: **Mar. 13, 2023**

Primary Examiner — Rakesh B Patel

(65) **Prior Publication Data**

US 2023/0327632 A1 Oct. 12, 2023

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(30) **Foreign Application Priority Data**

Mar. 17, 2022 (JP) 2022-043148

(57) **ABSTRACT**

(51) **Int. Cl.**

H03H 7/01 (2006.01)

H01F 27/28 (2006.01)

H01G 4/30 (2006.01)

H03H 1/00 (2006.01)

H03H 7/46 (2006.01)

A filter includes a multilayer body including dielectric layers and conductor layers, input and output terminals provided on a surface of the multilayer body, a first conductor pattern that has a planar shape forming at least a part of a closed pattern having an opening, a second conductor pattern that is connected to the input terminal, at least partially overlaps at least a part of the first conductor pattern, has an end portion located within the opening, and extends from an inside of the opening to an outside of the closed pattern, and a third conductor pattern that is connected to the output terminal, at least partially overlaps at least a part of the first conductor pattern, does not overlap the second conductor pattern, has an end portion located within the opening, and extends from the inside of the opening to the outside of the closed pattern.

(52) **U.S. Cl.**

CPC **H03H 7/0115** (2013.01); **H01F 27/2804** (2013.01); **H01G 4/30** (2013.01); **H03H 2001/0085** (2013.01); **H03H 7/46** (2013.01)

(58) **Field of Classification Search**

CPC H03H 2001/0085; H03H 7/0115

USPC 333/175, 185

See application file for complete search history.

12 Claims, 12 Drawing Sheets

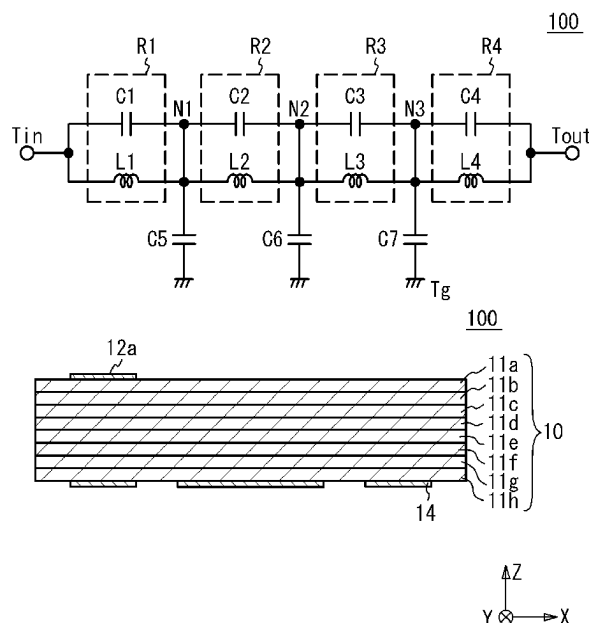


FIG. 1

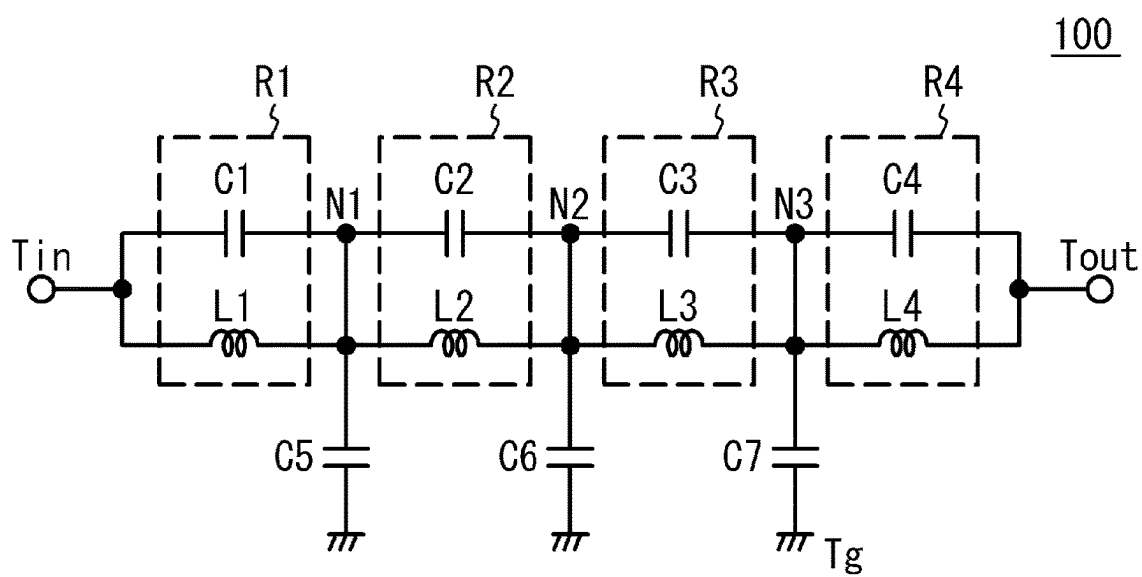


FIG. 2A

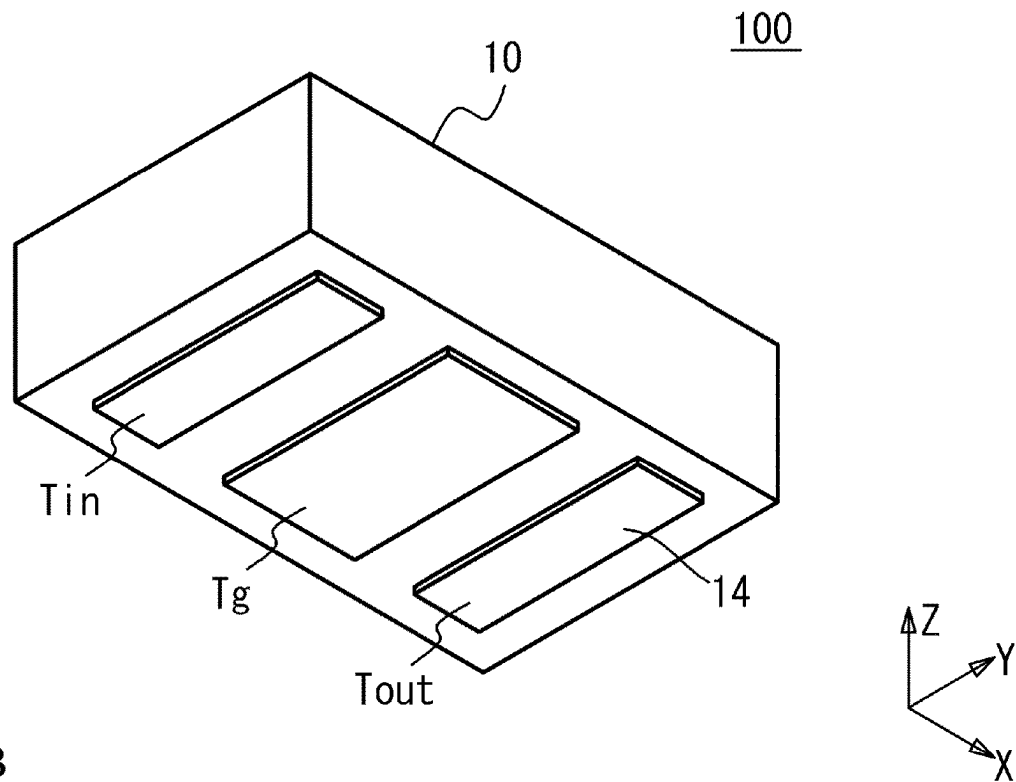


FIG. 2B

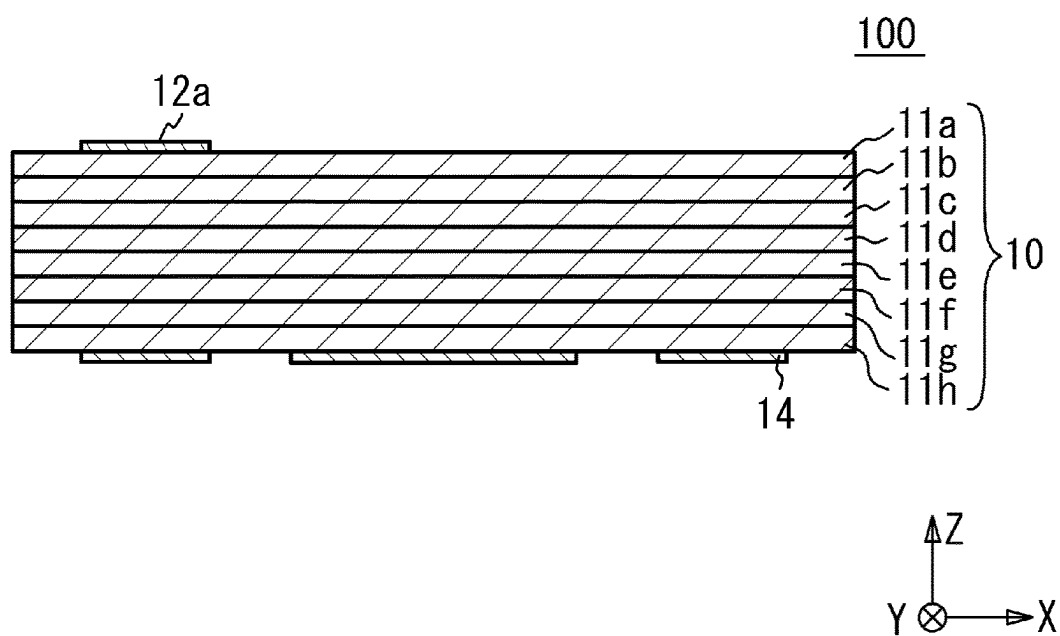


FIG. 3A

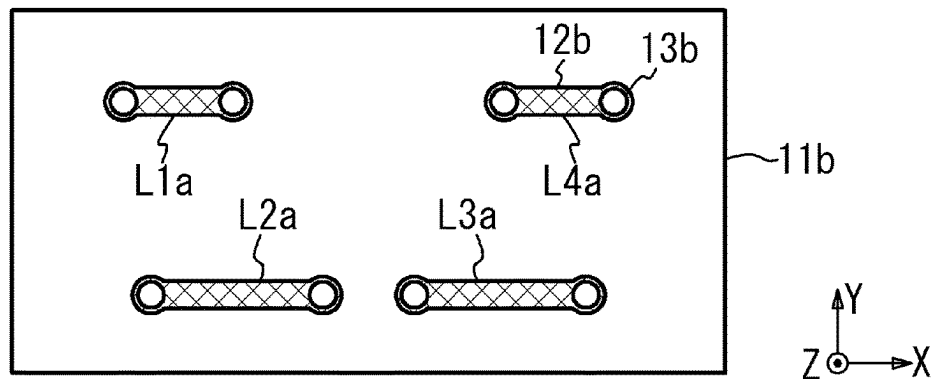


FIG. 3B

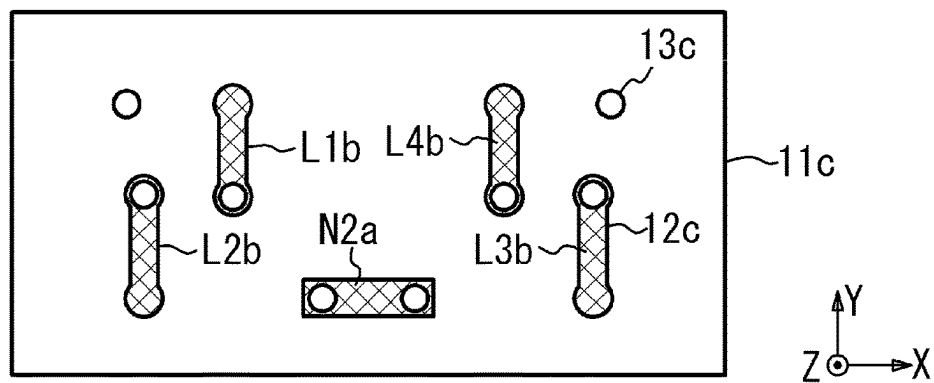


FIG. 3C

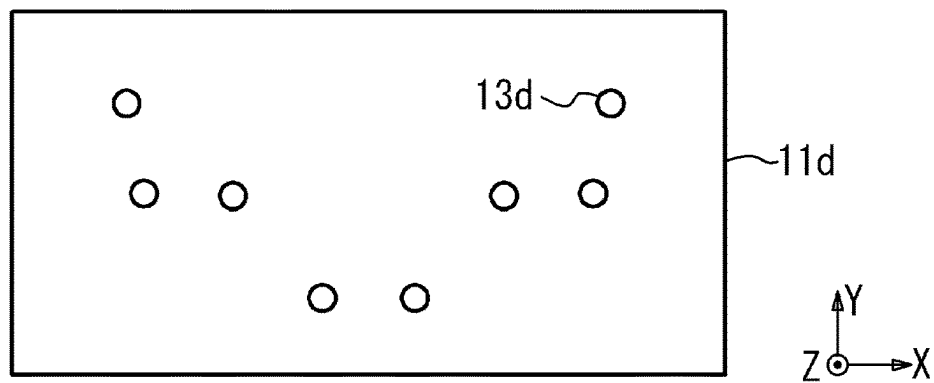


FIG. 3D

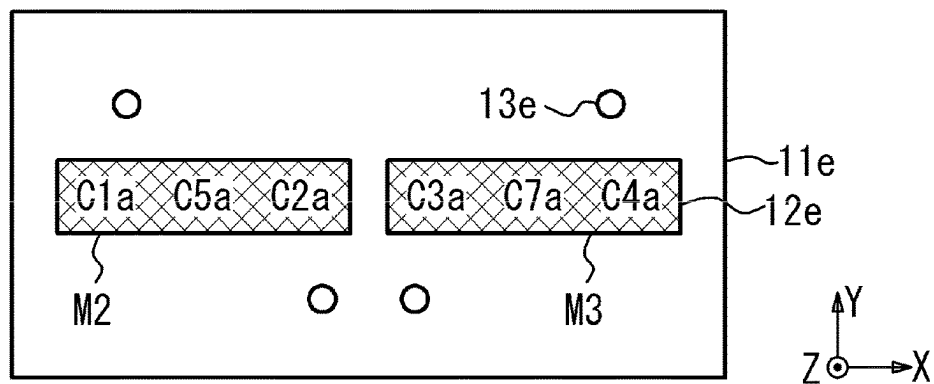


FIG. 4A

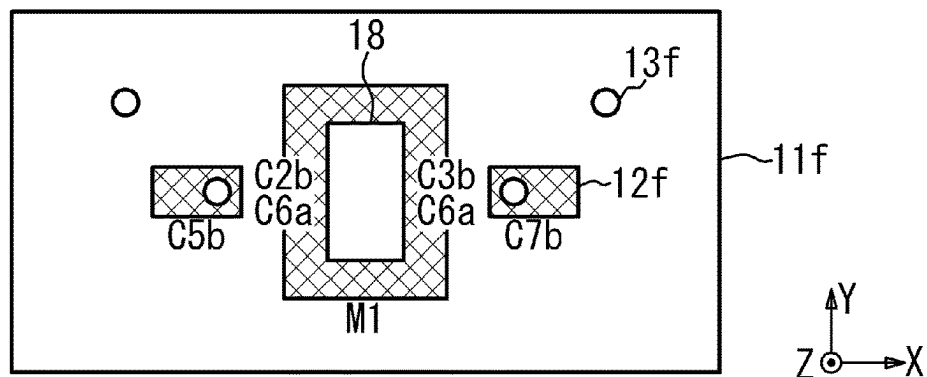


FIG. 4B

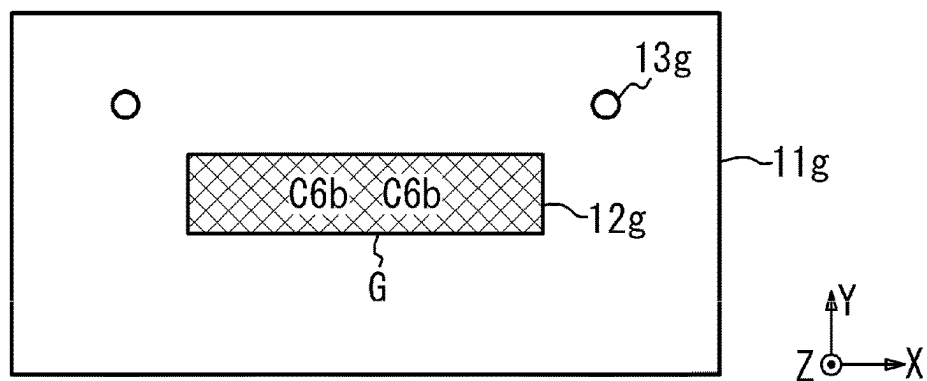


FIG. 4C

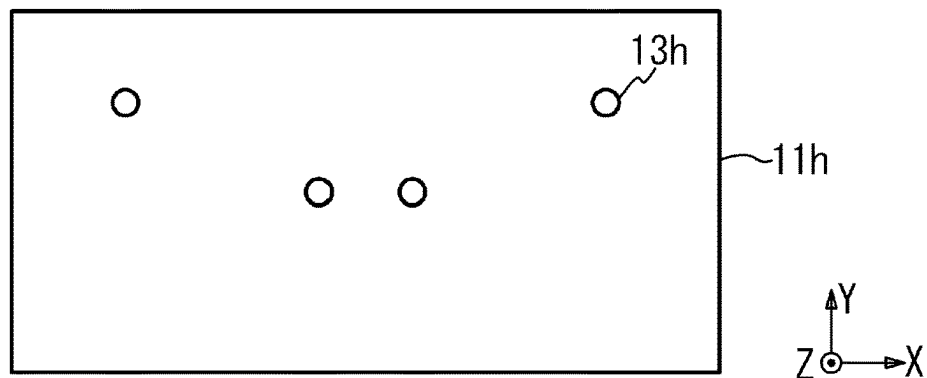


FIG. 4D

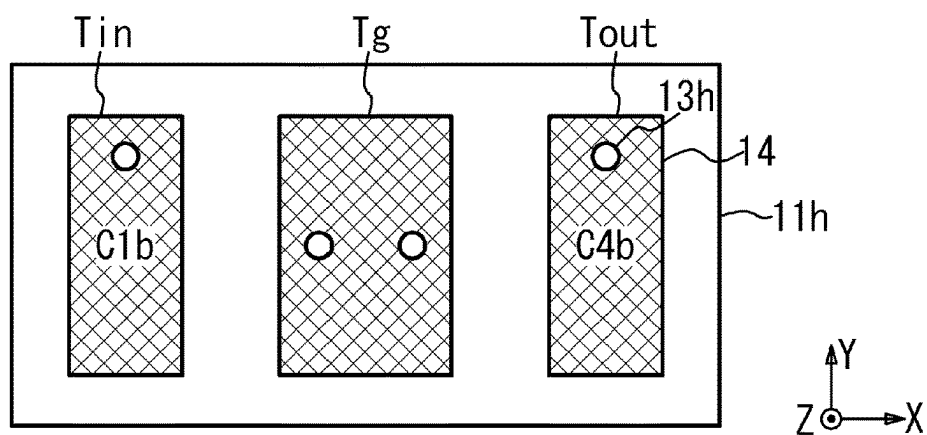


FIG. 5A

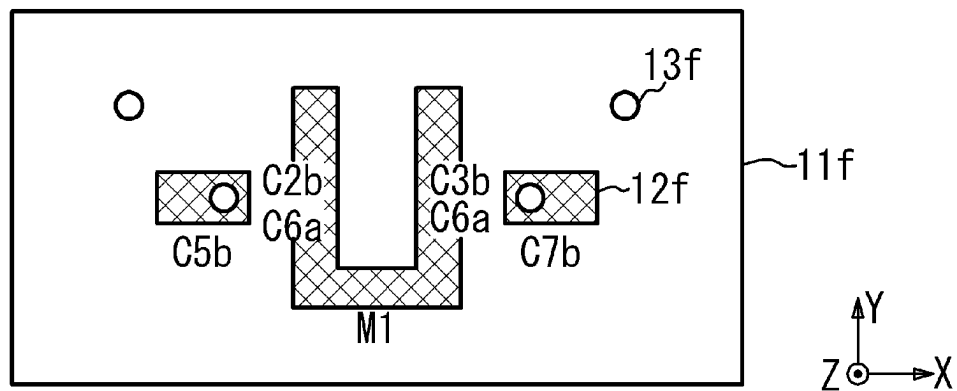


FIG. 5B

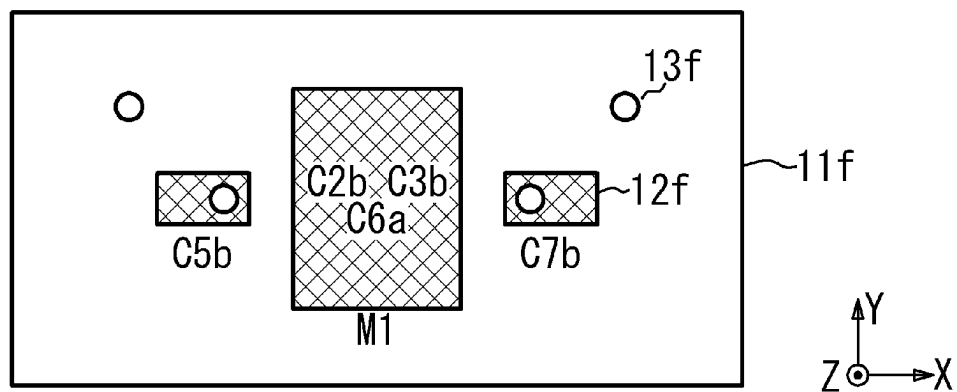


FIG. 6

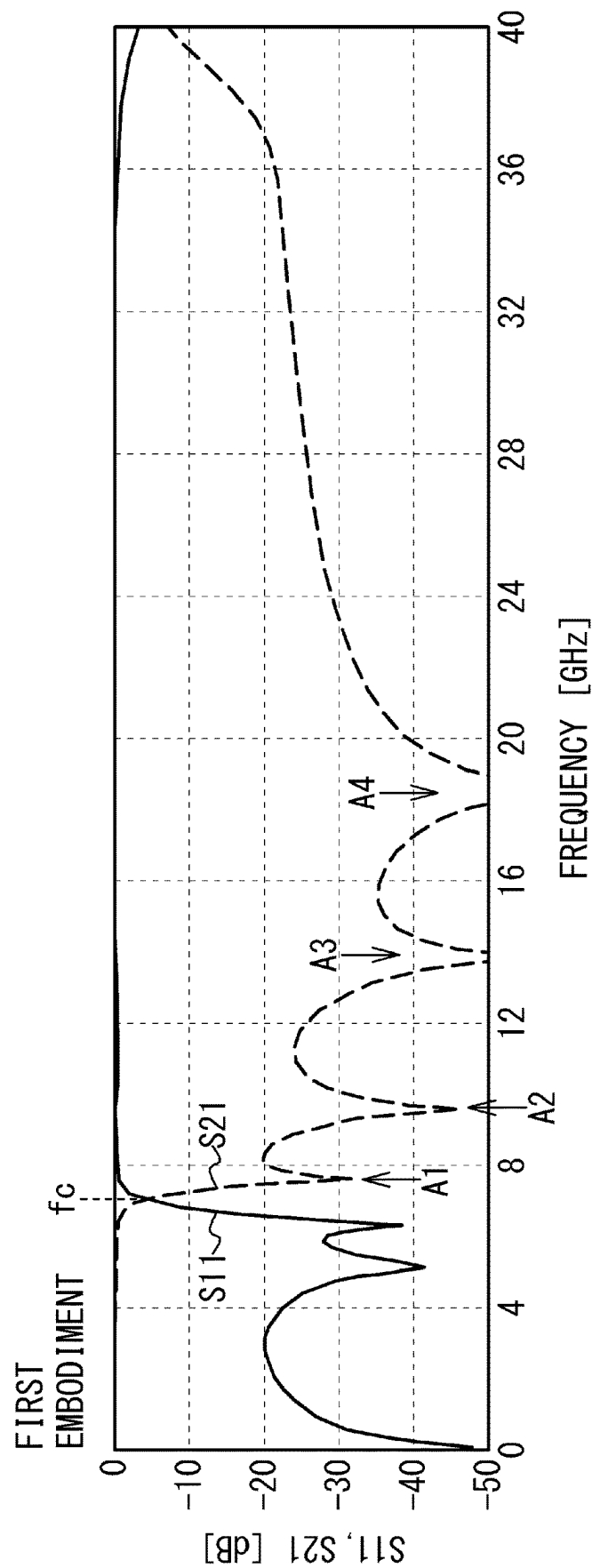


FIG. 7

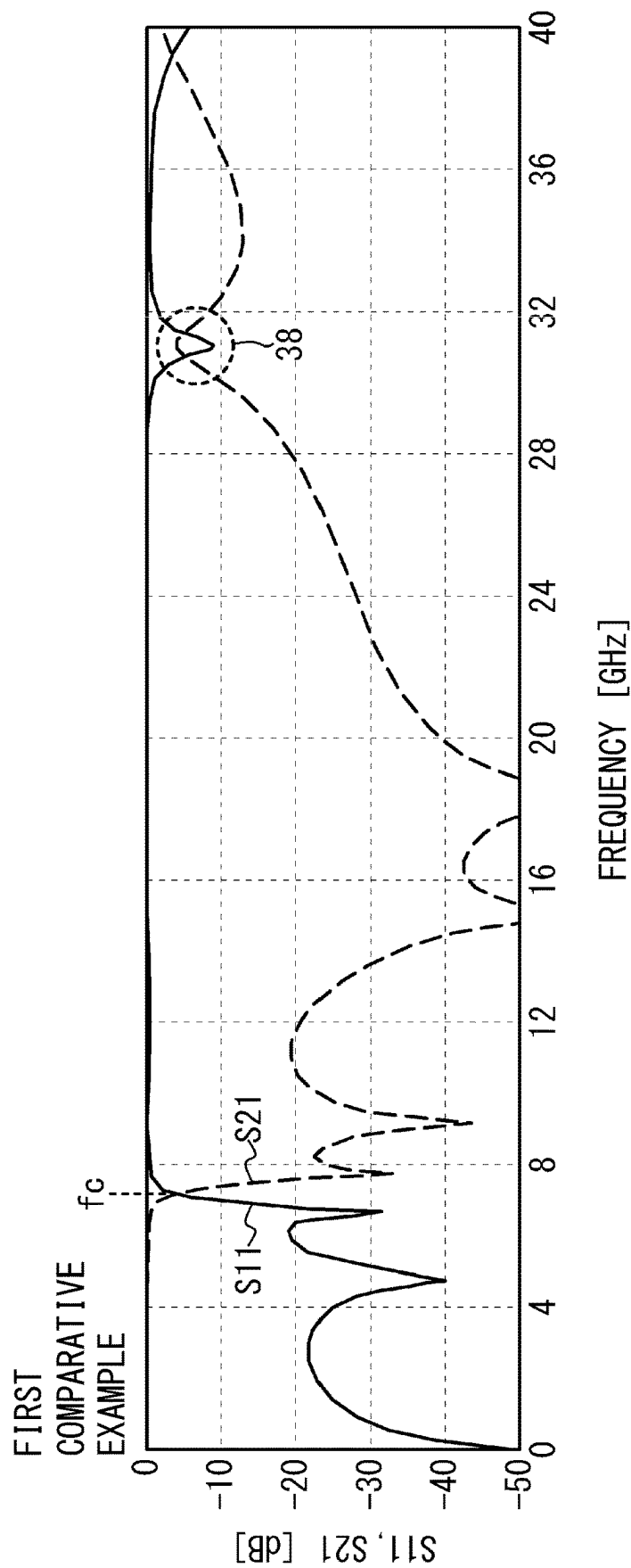


FIG. 8

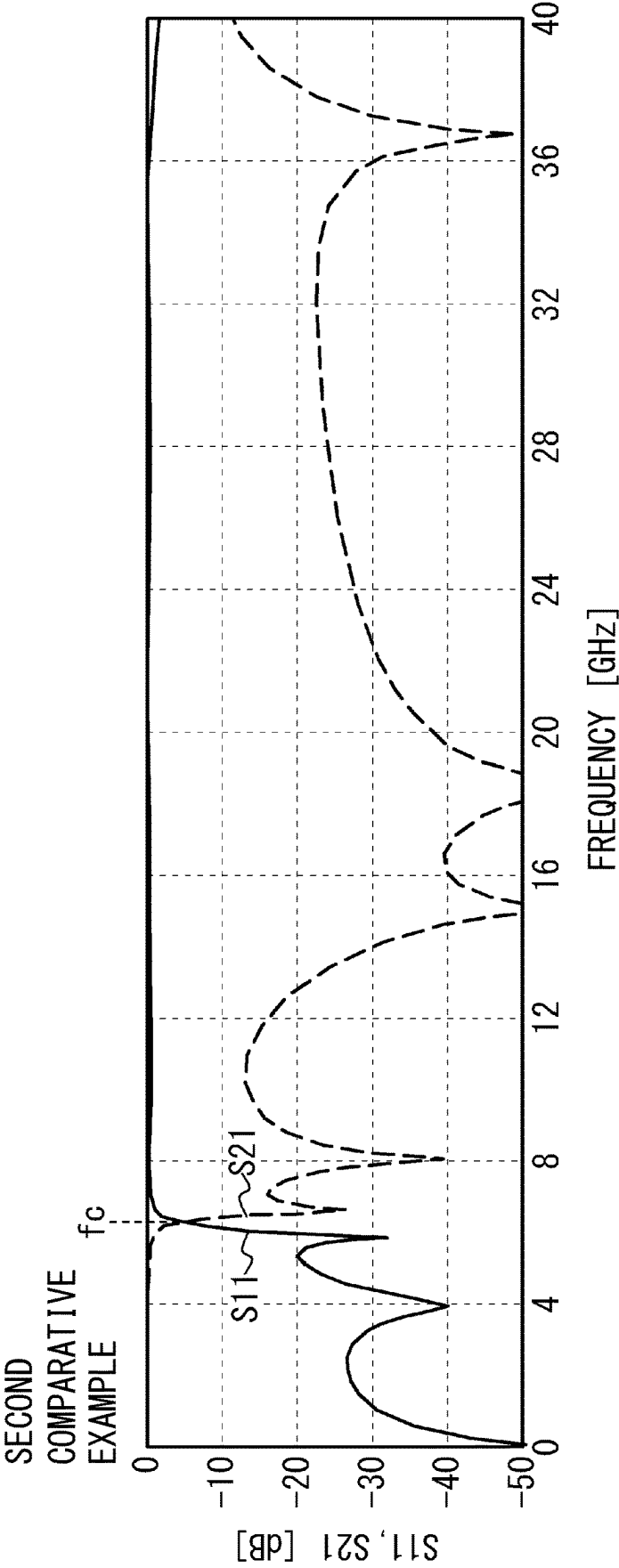


FIG. 9A

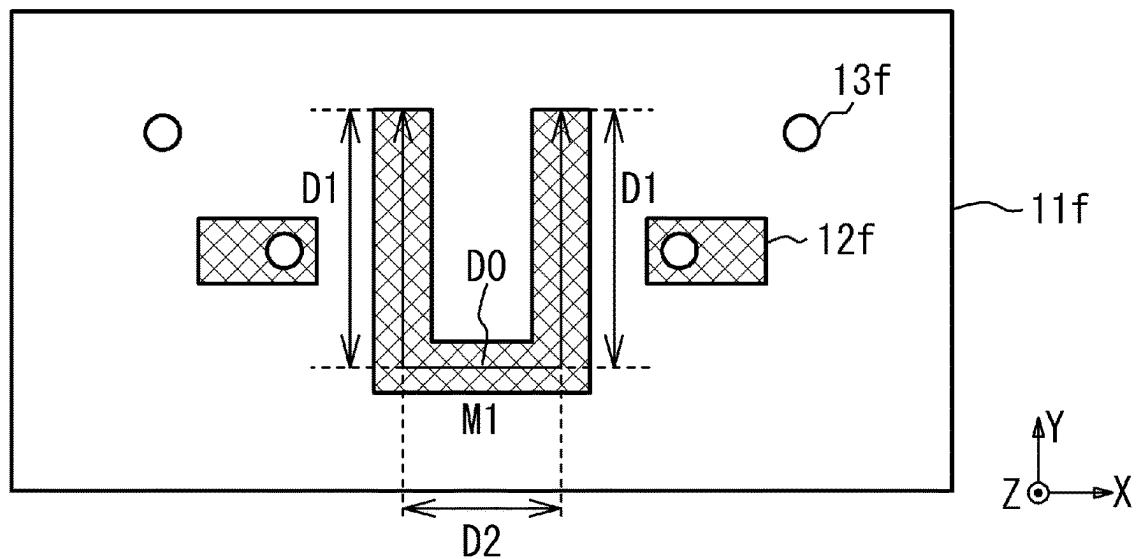


FIG. 9B

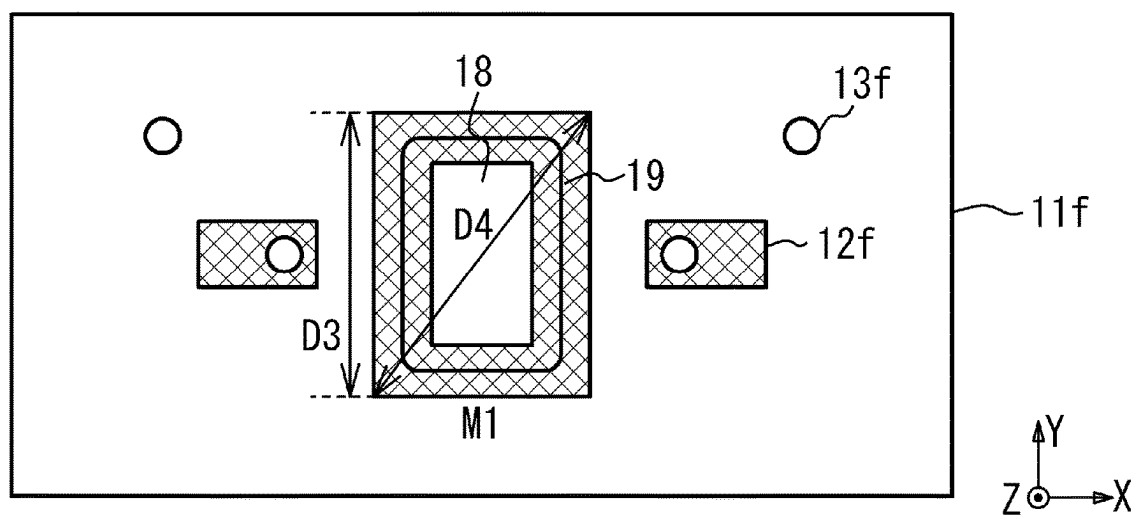


FIG. 10A

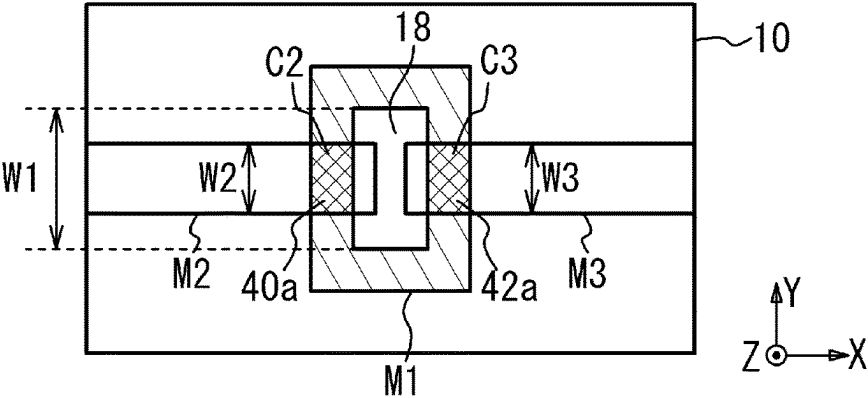


FIG. 10B

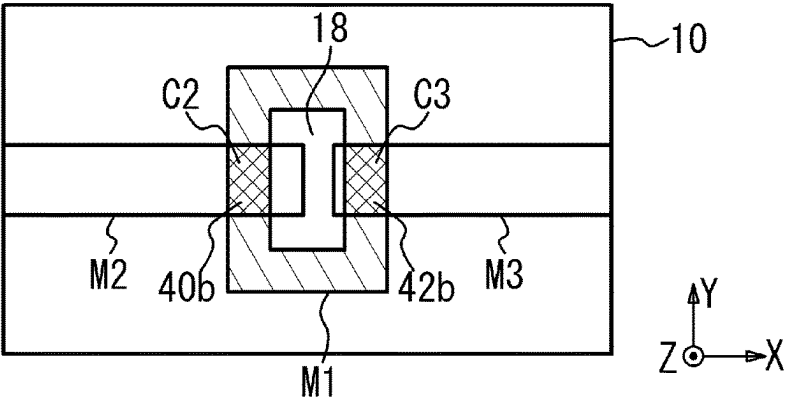


FIG. 10C

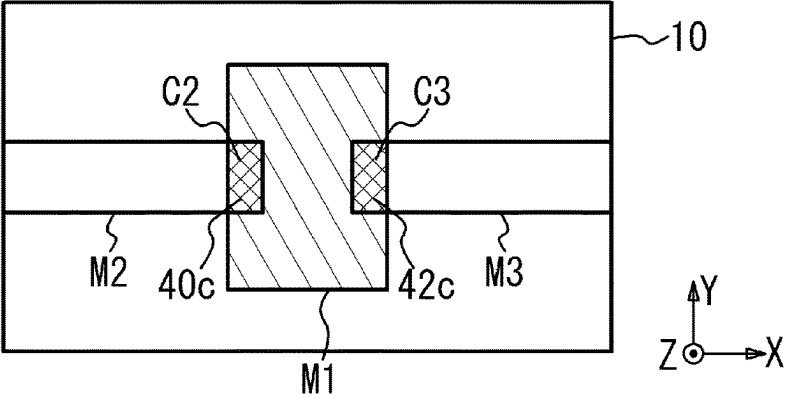


FIG. 10D

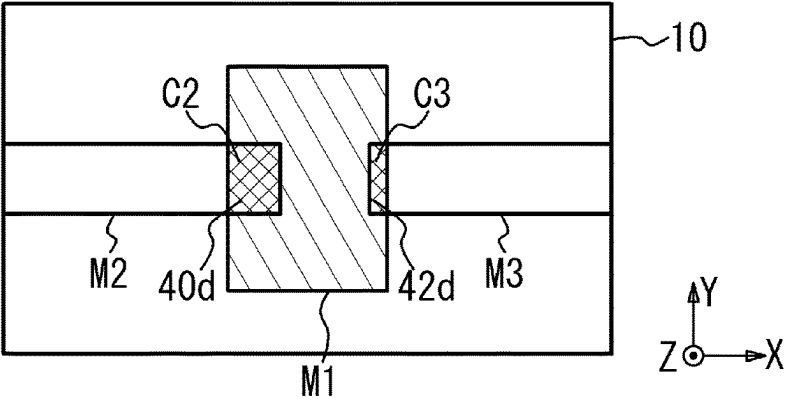


FIG. 11A

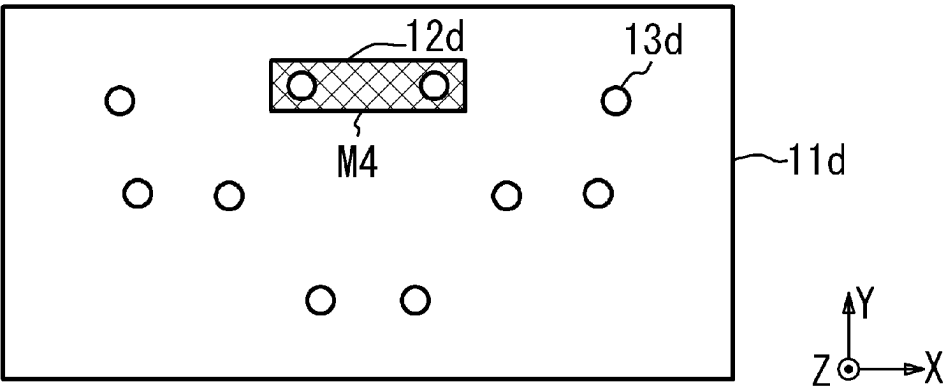


FIG. 11B

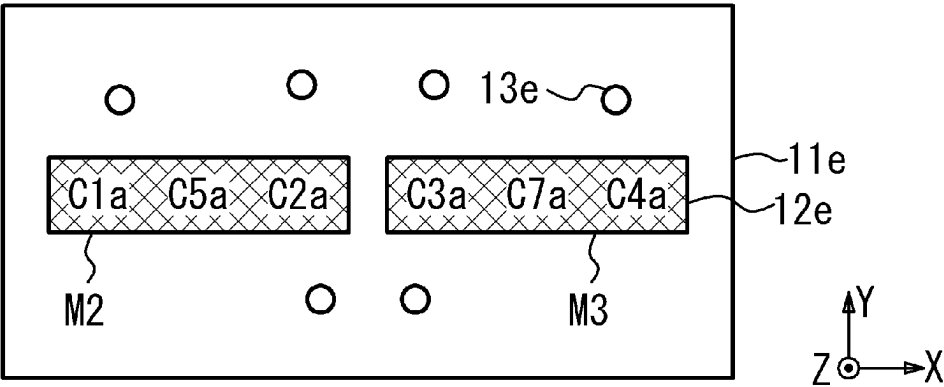


FIG. 11C

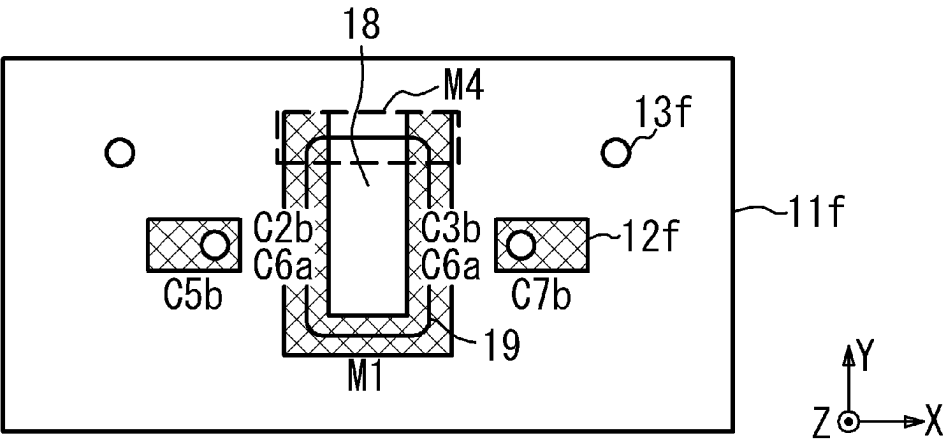
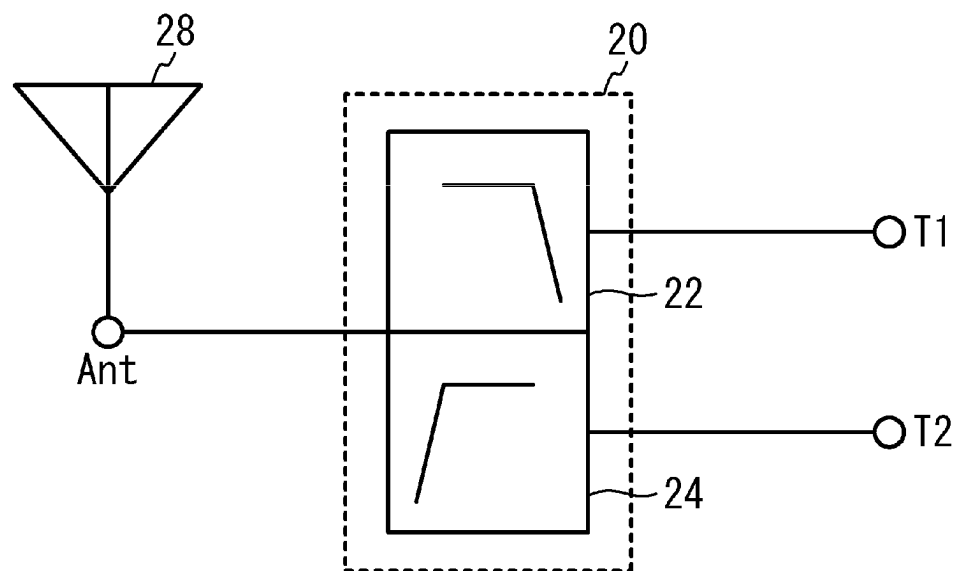


FIG. 12



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FILTER AND MULTIPLEXER**FIELD**

A certain aspect of the present disclosure relates to a filter and a multiplexer.

BACKGROUND

In mobile terminals such as notebook personal computers, smartphones, tablet terminals, or mobile phones, a filter that allows only high-frequency signals in a certain frequency band to pass therethrough is used. A multilayer filter in which dielectric layers are stacked is known as a filter, as disclosed in, for example, Japanese Patent Application Laid-Open Nos. 2013-219469 and 2018-19316 (Patent Documents 1 and 2).

RELATED ART DOCUMENTS**Patent Documents**

Japanese Patent Application Laid-Open No. 2013-219469
Japanese Patent Application Laid-Open No. 2018-19316

SUMMARY

In a filter for filtering high-frequency signals, it is required to reduce spurious emissions. An object of the present disclosure is to reduce spurious emissions.

In one aspect of the present disclosure, there is provided a filter including: a multilayer body in which a plurality of dielectric layers and a plurality of conductor layers are alternately stacked in a stacking direction; an input terminal provided on a surface of the multilayer body; an output terminal provided on a surface of the multilayer body; a first conductor pattern formed of a first conductor layer of the plurality of conductor layers, the first conductor pattern having a planar shape that forms at least a part of a closed pattern having an opening when viewed from the stacking direction; a second conductor pattern formed of a second conductor layer different from the first conductor layer of the plurality of conductor layers, and connected to the input terminal, at least a part of the second conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, an end portion of the second conductor pattern being located within the opening of the closed pattern when viewed from the stacking direction, the second conductor pattern extending from an inside of the opening to an outside of the closed pattern when viewed from the stacking direction; and a third conductor pattern formed of a third conductor layer that is same as or different from the second conductor layer and that is different from the first conductor layer of the plurality of conductor layers, and connected to the output terminal, at least a part of the third conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, the third conductor pattern not overlapping with the second conductor pattern when viewed from the stacking direction, an end portion of the third conductor pattern being located within the opening of the closed pattern when viewed from the stacking direction, the third conductor pattern extending from the inside of the opening to the outside of the closed pattern when viewed from the stacking direction.

In the above filter, the closed pattern may be a pattern in which a path through which a current flows is closed.

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In the above filter, the second conductor pattern and the third conductor pattern may be provided so as to face each other across the opening when viewed from the stacking direction.

In the above filter, a planar shape of a region where the first conductor pattern and the second conductor pattern overlap each other may be a substantially parallelogram when viewed from the stacking direction, and a planar shape of a region where the first conductor pattern and the third conductor pattern overlap each other may be a substantially parallelogram when viewed from the stacking direction.

In the above filter, the first conductor pattern and the second conductor pattern may form a first capacitor, and the first conductor pattern and the third conductor pattern may form a second capacitor.

The above filter may further include: a first inductor connected in parallel to the first capacitor between the input terminal and the output terminal; and a second inductor connected in parallel to the second capacitor between the input terminal and the output terminal.

In the above filter, a first parallel resonant circuit including the first capacitor and the first inductor may form an attenuation pole at a frequency higher than a passband, and a second parallel resonant circuit including the second capacitor and the second inductor may form an attenuation pole at a frequency higher than the passband.

The above filter may further include: a ground terminal provided on a surface of the multilayer body; and a third capacitor having a first end connected to the first conductor pattern and a second end connected to the ground terminal.

In the above filter, a first width, along the opening, of a region where the second conductor pattern overlaps the first conductor pattern when viewed from the stacking direction may be smaller than a width, in a width direction of the first width, of the opening, and a second width, along the opening, of a region where the third conductor pattern overlaps the first conductor pattern when viewed from the stacking direction may be smaller than a width, in a width direction of the second width, of the opening.

In the above filter, $X \leq 1.875 \times 10^7 / (f_c \times \sqrt{\epsilon_r})$ may be satisfied where X (m) is a maximum width of the closed pattern, f_c (Hz) is a high frequency end of a passband, and ϵ_r is a relative permittivity of each of the plurality of dielectric layers.

In the above filter, the closed pattern may be formed of the first conductive layer.

In the above filter, the closed pattern may include: the first conductor pattern, a fourth conductor pattern formed of a fourth conductor layer different from the first conductor layer of the plurality of conductor layers, and a via wiring that penetrates through one or some dielectric layers of the plurality of dielectric layers and connects the first conductor pattern and the fourth conductor pattern.

In the above filter, the filter may be a low-pass filter or a band-pass filter.

In another aspect of the present disclosure, there is provided a multiplexer including the above filter.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a circuit diagram of a filter in accordance with a first embodiment;

FIG. 2A and FIG. 2B are a perspective view and a cross-sectional view of the filter in accordance with the first embodiment, respectively;

FIG. 3A to FIG. 3D are plan views of dielectric layers in the first embodiment;

FIG. 4A to FIG. 4D are plan views of the dielectric layers in the first embodiment;

FIG. 5A and FIG. 5B are plan views of dielectric layers in first and second comparative examples, respectively;

FIG. 6 presents the transmission characteristics and reflection characteristics of the first embodiment in a simulation;

FIG. 7 presents the transmission characteristics and reflection characteristics of the first comparative example in the simulation;

FIG. 8 presents the transmission characteristics and reflection characteristics of the second comparative example in the simulation;

FIG. 9A and FIG. 9B are plan views of the dielectric layer 11f in the first comparative example and the first embodiment, respectively;

FIG. 10A and FIG. 10B are plan views of conductor patterns M1 to M3 in the first embodiment, and FIG. 10C and FIG. 10D are plan views of the conductor patterns M1 to M3 in the second comparative example;

FIG. 11A to FIG. 11C are plan views of the dielectric layers in a first variation of the first embodiment; and

FIG. 12 is a circuit diagram of a diplexer in accordance with a second embodiment.

DETAILED DESCRIPTION

Hereinafter, embodiments of the present disclosure will be described with reference to the drawings.

FIRST EMBODIMENT

As a first embodiment, a low-pass filter (LPF) used in a 5th generation (5G) mobile communication system will be described as an example.

In the 5G communication system, millimeter waves in the 28 GHz band or the like are used, and even a filter having a passband of 6 GHz or less is required to have improved attenuation characteristics in the frequency band from the passband to around 30 GHz.

FIG. 1 is a circuit diagram of a filter in accordance with the first embodiment. As illustrated in FIG. 1, a filter 100 of the first embodiment includes an input terminal T_{in} , an output terminal T_{out} , a ground terminal T_g , capacitors C1 to C7, and inductors L1 to L4. Parallel resonant circuits R1 to R4 are connected in series between the input terminal T_{in} and the output terminal T_{out} . A node between the parallel resonant circuits R1 and R2 is a node N1, a node between the parallel resonant circuits R2 and R3 is a node N2, and a node between the parallel resonant circuits R3 and R4 is a node N3. In the parallel resonant circuits R1 to R4, the capacitors C1 to C4 and the inductors L1 to L4 are connected in parallel, respectively. The capacitors C5 to C7 are connected between the nodes N1 to N3 and the ground terminal T_g , respectively.

FIG. 2A and FIG. 2B are a perspective view and a cross-sectional view of the filter in accordance with the first embodiment, respectively. A stacking direction of dielectric layers 11a to 11h is defined as a Z direction, an arrangement direction of terminals 14 in planar directions of the dielectric layers 11a to 11h is defined as an X direction, and a direction orthogonal to the X direction is defined as a Y direction. As illustrated in FIG. 2A and FIG. 2B, the filter 100 includes a multilayer body 10. The multilayer body 10 includes a plurality of the dielectric layers 11a to 11h that are stacked. The terminals 14 are provided on the lower surface of the multilayer body 10. The terminals 14 are, for example, the

input terminal T_{in} , the output terminal T_{out} , and the ground terminal T_g . On the upper surface of the multilayer body 10, a direction identification mark is provided by a conductor layer 12a.

FIG. 3A to FIG. 4D are plan views of the dielectric layers in the first embodiment. FIG. 3A to FIG. 4C illustrate conductor layers on the upper surfaces of the dielectric layers 11b to 11h and via wirings 13b to 13h penetrating through the dielectric layers 11b to 11h, respectively. FIG. 4D illustrates the terminals 14 and the via wirings 13h on the lower surface of the dielectric layer 11h as transparently viewed through the dielectric layer 11h from above. An electrode C1a in a conductor pattern M2 in FIG. 3D and an electrode C1b in the input terminal T_{in} in FIG. 4D indicate that the electrodes C1a and C1b form a capacitor C1 by sandwiching the dielectric layers 11e to 11h therebetween. The same applies to electrodes C2a to C7a and C2b to C7b in FIG. 3D to FIG. 4B and FIG. 4D. The same applies to the following similar drawings.

As illustrated in FIG. 3A, a conductor layer 12b forming line patterns L1a to L4a is provided on the upper surface of the dielectric layer 11b. As illustrated in FIG. 3B, a conductor layer 12c forming line patterns L1b to L4b and a conductor pattern N2a is provided on the upper surface of the dielectric layer 11c. The via wirings 13c penetrating through the dielectric layer 11c are provided.

As illustrated in FIG. 3C, no conductor layer is formed on the upper surface of the dielectric layer 11d. The via wirings 13d penetrating through the dielectric layer 11d are provided. As illustrated in FIG. 3D, on the upper surface of the dielectric layer 11e, the conductor layer 12e forming conductor patterns M2 and M3 is provided. The via wirings 13e penetrating through the dielectric layer 11e are provided. The conductor pattern M2 includes the electrodes C1a, C2a, and C5a, and corresponds to the node N1. The conductor pattern M3 includes the electrodes C3a, C4a, and C7a, and corresponds to the node N3.

As illustrated in FIG. 4A, a conductor layer 12f forming a conductor pattern M1 and the electrodes C5b and C7b is provided on the upper surface of the dielectric layer 11f. The via wirings 13f penetrating through the dielectric layer 11f are provided. The conductor pattern M1 includes the electrodes C2b, C3b, and C6a. The conductor patterns M1 and N2a are electrically connected to each other through the via wirings 13c to 13e, and correspond to the node N2.

As illustrated in FIG. 4B, a conductor layer 12g forming a conductor pattern G is provided on the upper surface of the dielectric layer 11g. The via wirings 13g penetrating through the dielectric layer 11g are provided. The conductor pattern G is a ground pattern electrically connected to the ground terminal T_g through the via wirings 13g and 13h, and includes the electrode C6b. The conductor pattern G is electrically connected to the electrodes C5b and C7b of FIG. 4A through the via wirings 13f.

As illustrated in FIG. 4C, no conductor pattern is formed on the upper surface of the dielectric layer 11h. The via wirings 13h penetrating through the dielectric layer 11h are provided. As illustrated in FIG. 4D, the terminals 14 are formed on the lower surface of the dielectric layer 11h. The terminals 14 include the input terminal T_{in} , the output terminal T_{out} , and the ground terminal T_g . The input terminal T_{in} includes the electrode C1b, and the output terminal T_{out} includes the electrode C4b.

The inductor L1 is formed of the line patterns L1a and L1b and the via wirings 13b to 13h, and is connected between the input terminal T_{in} and the conductor pattern M2 corresponding to the node N1. The inductor L2 is formed of

the line patterns **L2a** and **L2b** and the via wirings **13b** to **13d**, and is connected between the conductor pattern **M2** corresponding to the node **N1** and the conductor pattern **N2a** corresponding to the node **N2**. The inductor **L3** is formed of the line patterns **L3a** and **L3b** and the via wirings **13b** to **13d**, and is connected between the conductor pattern **N2a** corresponding to the node **N2** and the conductor pattern **M3** corresponding to the node **N3**. The inductor **L4** is formed of the line patterns **L4a** and **L4b** and the via wirings **13b** to **13h**, and is connected between the conductor pattern **M3** corresponding to the node **N3** and the output terminal **Tout**.

The capacitor **C1** is formed of the electrodes **C1a** and **C1b** sandwiching the dielectric layers **11e** to **11h** therebetween, and is connected between the input node **Tin** and the conductor pattern **M2** corresponding to the node **N1**. The capacitor **C2** is formed of the electrodes **C2a** and **C2b** sandwiching the dielectric layer **11e** therebetween, and is connected between the conductor pattern **M2** corresponding to the node **N1** and the conductor pattern **M1** corresponding to the node **N2**. The capacitor **C3** is formed of the electrodes **C3a** and **C3b** sandwiching the dielectric layer **11e** therebetween, and is connected between the conductor pattern **M1** corresponding to the node **N2** and the conductor pattern **M3** corresponding to the node **N3**. The capacitor **C4** is formed of the electrodes **C4a** and **C4b** sandwiching the dielectric layers **11e** to **11h** therebetween, and is connected between the conductor pattern **M3** corresponding to the node **N3** and the output terminal **Tout**.

The capacitor **C5** is formed of the electrodes **C5a** and **C5b** sandwiching the dielectric layer **11e** therebetween, and is connected between the conductor pattern **M2** corresponding to the node **N1** and the ground terminal **Tg**. The capacitor **C6** is formed of the electrodes **C6a** and **C6b** sandwiching the dielectric layer **11f** therebetween, and is connected between the conductor pattern **M1** corresponding to the node **N2** and the ground terminal **Tg**. The capacitor **C7** is formed of the electrodes **C7a** and **C7b** sandwiching the dielectric layer **11e** therebetween, and is connected between the conductor pattern **M3** corresponding to the node **N3** and the ground terminal **Tg**.

The dielectric layers **11a** to **11h** are made of a ceramic material and contain, for example, an oxide of Si, Ca, and Mg (for example, $\text{CaMgSi}_2\text{O}_6$, which is a diopside crystal) as a main component. The main components of the dielectric layers **11a** to **11h** may be oxides other than oxides of Si, Ca and/or Mg. Furthermore, the dielectric layers **11a** to **11h** may contain at least one of oxides of Ti, Zr, and Al as an insulating material.

The conductor layers **12a** to **12c** and **12e** to **12g**, the via wirings **13b** to **13h**, and the upper portions of the terminals **14** are metal layers containing, for example, Ag, Pd, Pt, Cu, Ni, Au, Au—Pd alloys, or Ag—Pt alloys as a main component. The upper portion of the terminal **14** may contain a non-conductive material such as TiO_2 , ZrO_2 , or Al_2O_3 in addition to the above metal material. The lower portion of the terminal **14** is a Ni film and a Sn film.

In the first embodiment, as illustrated in FIG. 4A, the planar shape of the conductor pattern **M1** is a quadrangular shape, and the conductor pattern **M1** has an opening **18** having a quadrangular planar shape. Thus, the conductor pattern **M1** is a closed pattern in which a path through which a current flows is closed.

COMPARATIVE EXAMPLES

FIG. 5A and FIG. 5B are plan views of dielectric layers in first and second comparative examples, respectively. As

illustrated in FIG. 5A, in the first comparative example, the planar shape of the conductor pattern **M1** on the upper surface of the dielectric layer **11f** is a U shape obtained by removing one side from the sides of a quadrangular shape, and is not a closed pattern. As illustrated in FIG. 5B, in the second comparative example, the planar shape of the conductor pattern **M1** in the dielectric layer **11f** is a solid quadrangular shape without an opening. The upper surfaces of other dielectric layers **11b** to **11e**, **11g**, and **11h** and the lower surface of the dielectric layer **11h** in the first and second comparative examples have the same configurations as those in FIG. 3A to FIG. 3D and FIG. 4B to FIG. 4D, respectively.

Simulation

Three-dimensional electromagnetic field simulation was performed on the first embodiment and the first and second comparative examples using a finite element method to simulate the pass characteristics and reflection characteristics of the filter.

Each of the dielectric layers **11a** to **11h** contains $\text{CaMgSi}_2\text{O}_6$ as a main component and has a relative permittivity of about 10. Since the frequencies supported by the filter **100** are as high as 1 GHz to 40 GHz, the filter **100** functions like a distributed constant circuit. Therefore, although the capacitances of the capacitors **C1** to **C7** and the inductances of the inductors **L1** to **L4** are not determined, approximate values of the capacitances of the capacitors **C1** to **C7** and the inductances of the inductors **L1** to **L4** in the first embodiment and the first and second comparative examples are presented in Table 1.

TABLE 1

Capacitance [pF]							Inductance [nH]			
C1	C2	C3	C4	C5	C6	C7	L1	L2	L3	L4
0.1	1.4	1.4	0.1	0.45	0.35	0.45	0.8	0.4	0.4	0.8

FIG. 6 to FIG. 8 present the transmission characteristics and reflection characteristics of the first embodiment and the first and second comparative example in the simulation, respectively. The absolute value of **S21** of the S parameters indicates the transmission characteristics and the absolute value of **S11** of the S parameters indicates the reflection characteristics.

As presented in FIG. 6, when the frequency at which **S21** is -3 dB is referred to as a cutoff frequency f_c (i.e., the high frequency end of the passband), the cutoff frequency f_c is approximately 7 GHz. At frequencies lower than the cutoff frequency f_c , **S21** is approximately 0 dB and **S11** is -20 dB or less. At frequencies higher than the cutoff frequency f_c , **S21** is -20 dB or less, and **S11** is approximately 0 dB. The passband is a band equal to or less than the cutoff frequency f_c , and the attenuation band is from the cutoff frequency f_c to about 40 GHz.

Attenuation poles **A1** to **A4** in the attenuation band are mainly formed by the parallel resonant circuits **R1** to **R4**. Good attenuation characteristics are obtained around 30 GHz.

As presented in FIG. 7, in the first comparative example, the cutoff frequency f_c is about 7 GHz, which is the same as in the first embodiment. As indicated by a broken-line circle **38**, a spurious emission in which **S11** becomes smaller and **S21** becomes larger is observed around 31 GHz. Thus, the attenuation characteristics are deteriorated around 30 GHz.

As presented in FIG. 8, in the second comparative example, the spurious emission around 31 GHz observed in the first comparative example is not observed. The cutoff frequency f_c is about 7 GHz as in the first embodiment. Good attenuation characteristics are obtained around 30 GHz. As described above, in the first comparative example in which the planar shape of the conductor pattern M1 is a U shape, a spurious emission is generated around 31 GHz. The spurious emission is considered to be related to the planar shape of the conductor pattern M1.

In the first comparative example, the cause of the occurrence of the spurious emission around 31 GHz will be considered. FIG. 9A and FIG. 9B are plan views of the dielectric layer 11f in the first comparative example and the first embodiment, respectively. As illustrated in FIG. 9A, in the first comparative example, the conductor pattern M1 is U-shaped. Therefore, the conductor pattern M1 becomes a line having a length D0. The length D0 is the length of the center line of the conductor pattern M1. The length D0 is expressed by $D0=D1+D2+D1$. Here, $D1=375\text{ }\mu\text{m}$, $D2=275\text{ }\mu\text{m}$, and $D0=1025\text{ }\mu\text{m}$. When the length D0 is $\frac{1}{4}$ of the wavelength λ of the high-frequency signal, the conductor pattern M1 causes resonance. Therefore, the frequency at which the length D0 is $\lambda/4$ is calculated. When the relative permittivity of the multilayer body 10 is ϵ_r , the wavelength λ at which $D0=\lambda/4$ is $\lambda=4 \times D0 \times \sqrt{\epsilon_r}=4 \times 1.025 \times 10^{-3} \text{ m} \times \sqrt{10} \approx 1.3 \times 10^{-2} \text{ m}=13\text{ mm}$. When the speed of light is $c=3 \times 10^8 \text{ m/s}$, the frequency f_s at which the wavelength is λ is $f_s=c/\lambda \approx 23\text{ GHz}$. As described above, the frequency at which the conductor pattern M1 resonates when the length of the conductor pattern M1 is $\lambda/4$ has a numerical value close to 31 GHz at which the spurious emission occurs in the first comparative example.

As presented illustrated in FIG. 9B, in the first embodiment, the conductor pattern M1 has the opening 18 and is a closed pattern in which a current path 19 is closed. Therefore, a long line as in the first comparative example in FIG. 9A is not formed. The length D3 of the long side of the conductor pattern M1 is $475\text{ }\mu\text{m}$. The wavelength λ at which $D3=\lambda/4$ is $\lambda=4 \times D3 \times \sqrt{\epsilon_r}=4 \times 0.475 \times 10^{-3} \times \sqrt{10} \approx 6 \times 10^{-3} \text{ m}=6\text{ mm}$, and the frequency f_s is $f_s=c/\lambda \sim 50\text{ GHz}$. Therefore, the frequency at which the length of the conductor pattern M1 is $\lambda/4$ is 50 GHz. The length D4 of the diagonal line of the conductor pattern M1 is $605\text{ }\mu\text{m}$. The wavelength λ at which $D4=\lambda/4$ is $\lambda=4 \times D4 \times \sqrt{\epsilon_r}=4 \times 0.605 \times 10^{-3} \times \sqrt{10} \approx 7.7 \times 10^{-3} \text{ m}=7.7\text{ mm}$, and the frequency f_s is $f_s=c/\lambda \approx 40\text{ GHz}$. As described above, the frequency at which the length of the conductor pattern M1 is $\lambda/4$ is 40 to 50 GHz.

In the first comparative example, since the frequency at which the conductor pattern M1 resonates when the length of the conductor pattern is $\lambda/4$ is around 30 GHz, it is considered that a spurious emission is generated around 30 GHz as illustrated in FIG. 7. On the other hand, in the first embodiment and the second comparative example, since the frequency at which the conductor pattern M1 resonates is equal to or higher than 40 GHz, it is considered that no spurious emission is generated in FIG. 6 and FIG. 8.

FIG. 10A and FIG. 10B are plan views of the conductor patterns M1 to M3 in the first embodiment, and FIG. 10C and FIG. 10D are plan views of the conductor patterns M1 to M3 in the second comparative example.

As illustrated in FIG. 10A, in the first embodiment, when viewed from the Z direction, the end portion of the conductor pattern M2 is located within the opening 18 of the conductor pattern M1, and the conductor pattern M2 extends from the inside of the opening 18 to the outside of the conductor pattern M1. The end portion of the conductor

pattern M3 is located within the opening 18 of the conductor pattern M1, and the conductor pattern M3 extends from the inside of the opening 18 to the outside of the conductor pattern M1. The capacitor C2 is formed in a region 40a where the conductor patterns M1 and M2 overlap each other. The capacitor C3 is formed in a region 42a where the conductor patterns M1 and M3 overlap each other.

As illustrated in FIG. 10B, when the conductor pattern M1 is misaligned with the conductor patterns M2 and M3, for example, when the conductor patterns M2 and M3 are shifted from the conductor pattern M1 in the +X direction, the area of a region 40b where the conductor patterns M1 and M2 overlap each other and the area of a region 42b where the conductor patterns M1 and M2 overlap each other are almost the same as the area of the region 40a and the area of the region 42a in FIG. 10A, respectively. Therefore, even when misalignment occurs between the conductor patterns M2 and M3 and the conductor pattern M1, the capacitances of the capacitors C2 and C3 are almost the same as those in FIG. 10A.

As illustrated in FIG. 10C, in the second comparative example, the conductor pattern M1 is a solid pattern without the opening 18. The capacitor C2 is formed in a region 40c where the conductor patterns M1 and M2 overlap each other. The capacitor C3 is formed in a region 42c where the conductor patterns M1 and M3 overlap each other.

As illustrated in FIG. 10D, for example, when the conductor patterns M2 and M3 are shifted from the conductor pattern M1 in the +X direction, the area of a region 40d where the conductor patterns M1 and M2 overlap becomes larger than the area of the region 40c in FIG. 10C, and the area of a region 42d where the conductor patterns M1 and M3 overlap becomes smaller than the area of the region 42c in FIG. 10C. As a result, the capacitance of the capacitor C2 becomes larger than that in FIG. 10C, and the capacitance of the capacitor C3 becomes smaller than that in FIG. 10C. When a filter is designed using the capacitances of the capacitors C2 and C3 illustrated in FIG. 10C, in FIG. 10D, the capacitances of the capacitors C2 and C3 deviate from the design values.

As described above, in the first embodiment, by providing the conductor patterns M2 and M3 so as to cross the ring-shaped closed pattern of the conductor pattern M1 when viewed from the Z direction, even when misalignment occurs between the conductor pattern M1 and the conductor patterns M2 and M3, it is possible to reduce a change in capacitance of the capacitor C2 and a change in capacitance of the capacitor C3.

First Variation of the First Embodiment

FIG. 11A to FIG. 11C are plan views of dielectric layers in a first variation of the first embodiment. In FIG. 11C, a conductor pattern M4 viewed from the Z direction is indicated by a broken line.

As illustrated in FIG. 11A, the conductor layer 12d forming the conductor pattern M4 is provided on the upper surface of the dielectric layer 11d. As illustrated in FIG. 11B, the dielectric layer 11e is provided with the via wirings 13e connected to the conductor pattern M4, unlike in FIG. 3D. As illustrated in FIG. 11C, the conductor pattern M1 is U-shaped. As illustrated in FIG. 11A and FIG. 11C, the planar shape formed of the combination of the conductor patterns M1 and M4 as viewed from the Z direction is a closed pattern. A first end portion of the conductor pattern M1 and a first end portion of the conductor pattern M4 are electrically connected to each other through the via wirings

13d and 13e, and a second end portion of the conductor pattern M1 and a second end portion of the conductor pattern M4 are electrically connected to each other through the via wirings 13d and 13e. As a result, the closed loop current path 19 is formed in a closed pattern.

As in the first variation of the first embodiment, even when the conductor pattern M1 forming the electrodes C2b and C3b of the capacitors C2 and C3 is U-shaped, it is only necessary to form a closed pattern together with another conductor pattern M4.

In the first embodiment and the variation thereof, in the multilayer body 10, the conductor pattern M1 (a first conductor pattern) whose planar shape when viewed from the Z direction forms at least a part of the closed pattern having the opening 18 is formed of the conductor layer 12f (a first conductor layer). The closed pattern is a pattern in which the path 19 through which a current flows is closed. This configuration increases the frequency at which the conductor pattern M1 resonates when the length thereof is $\lambda/4$, and causes spurious emissions to be less likely to occur in the attenuation band.

The conductor pattern M2 (a second conductor pattern) is formed of the conductor layer 12e (a second conductor layer) different from the conductor layer 12f, and is connected to the input terminal Tin. When viewed from the Z direction, at least a part of the conductor pattern M2 overlaps at least a part of the conductor pattern M1 to form the capacitor C2 (a first capacitor). The conductor pattern M3 (a third conductor pattern) is formed of the conductor layer 12e (a third conductor layer) different from the conductor layer 12f and is connected to the output terminal Tout. At least a part of the conductor pattern M3 overlaps at least a part of the conductor pattern M1 to form the capacitor C3 (a second capacitor). When viewed from the Z direction, an end portion of the conductor pattern M2 is located within the opening 18, and the conductor pattern M2 extends from the inside of the opening 18 to the outside of the closed pattern. When viewed from the Z direction, the conductor pattern M3 does not overlap with the conductor pattern M2, and an end portion of the conductor pattern M3 is located within the opening 18, and the conductor pattern M3 extends from the inside of the opening 18 to the outside of the closed pattern. Thus, as illustrated in FIG. 10A and FIG. 10B, even when misalignment occurs between the conductor pattern M1 and the conductor patterns M2 and M3, it is possible to inhibit the capacitances of the capacitors C2 and C3 from deviating from the design values.

As illustrated in FIG. 10A and FIG. 10B, when viewed from the Z direction, the conductor patterns M2 and M3 are provided so as to face each other across the opening 18. Thus, even when misalignment occurs between the conductor pattern M1 and the conductor patterns M2 and M3, it is possible to further inhibit the capacitances of the capacitors C2 and C3 from deviating from the design value.

In the first embodiment and the first variation thereof, the planar shape of the closed pattern is a quadrangular shape. However, the planar shape of the closed pattern may be an O-shape such as a circular shape or an elliptical shape. However, when viewed from the Z direction, the planar shapes of the regions 40a and 40b where the conductor patterns M1 and M2 overlap are preferably substantially parallelograms to the extent of manufacturing errors, and the planar shapes of the regions 42a and 42b where the conductor patterns M1 and M3 overlap are preferably substantially parallelograms to the extent of manufacturing errors. This configuration allows the area of the region 40a and the area of the region 40b to be almost the same, and the area

of the region 42a and the area of the region 42b to be almost the same even when misalignment occurs between the conductor pattern M1 and the conductor patterns M2 and M3. Therefore, it is possible to further inhibit the capacitances of the capacitors C2 and C3 from deviating from the design values.

The first width W2, along the opening 18, of the region 40a where the conductor pattern M2 overlaps the conductor pattern M1 when viewed from the Z direction is smaller than the width W1, in the width direction of the width W2, of the opening 18. The second width W3, along the opening 18, of the region 42a where the conductor pattern M3 overlaps the conductor pattern M1 when viewed from the Z direction is smaller than the width W1, in the width direction of the width W3, of the opening 18. This configuration allows the area of the region 40a and the area of the region 40b to be almost the same, and the area of the region 42a and the area of the region 42b to be almost the same even when misalignment occurs between the conductor pattern M1 and the conductor patterns M2 and M3. Therefore, it is possible to further inhibit the capacitances of the capacitors C2 and C3 from deviating from the design values. The widths W2 and W3 are preferably equal to or less than 0.9 times the width W1, and more preferably equal to or less than 0.8 times the width W1.

As illustrated in FIG. 1, between the input terminal Tin and the output terminal Tout, the inductor L2 (a first inductor) is connected in parallel to the capacitor C2, and the inductor L3 (a second inductor) is connected in parallel to the capacitor C3. In such a configuration, resonance caused by a $\lambda/4$ -line becomes a problem. Therefore, it is preferable to form the electrodes C2b and C3b in a closed pattern.

The parallel resonant circuit R2 (a first parallel resonant circuit) including the capacitor C2 and the inductor L2 forms the attenuation pole at a frequency higher than the passband, and the parallel resonant circuit R3 (a second parallel resonant circuit) including the capacitor C3 and the inductor L3 forms the attenuation pole at a frequency higher than the passband. In such a configuration, resonance caused by a $\lambda/4$ -line becomes a problem. Therefore, it is preferable to form the electrodes C2b and C3b in a closed pattern.

A first end of the capacitor C6 (a third capacitor) is connected to the conductor pattern M1 and a second end thereof is connected to the ground terminal Tg. In such a configuration, resonance caused by a $\lambda/4$ -line becomes a problem. Therefore, it is preferable to form the electrodes C2b and C3b in a closed pattern.

In the first embodiment and the first variation thereof, only two parallel resonant circuits R2 and R3 of the parallel resonant circuits R1 and R4 may be provided, or three or five or more parallel resonant circuits may be provided. In the filter 100, the input terminal Tin and the output terminal Tout are symmetric, that is, the capacitances of the capacitors are symmetric with respect to the node N2 and the capacitor C6, and the inductances of the inductors are symmetric with respect to the node N2 and the capacitor C6. The input terminal Tin and the output terminal Tout may be asymmetric.

In the 5G communication system, signals of 6 GHz or less and signals of 28 GHz are used. Therefore, when the high frequency end of the passband (i.e., the cutoff frequency fc at which S21 is -3 dB) is 7 GHz, the frequency of the spurious emission is preferably 28 GHz or higher. Therefore, the frequency of the spurious emission caused by the resonance of the $\lambda/4$ -line of the closed pattern is preferably equal to or greater than four times the cutoff frequency fc. Therefore, when the maximum width of the closed pattern is X

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(m), the cutoff frequency is f_c (Hz), and the relative permittivity of each of the plurality of the dielectric layers **11a** to **11h** is ϵ_r , the frequency f_r of the spurious emission caused by the resonance of the $\lambda/4$ -line is $f_r = 4 \times f_c \geq c / (4 \times X \times \sqrt{\epsilon_r})$. Therefore, it is preferable that $X \leq c / (4 \times 4 \times f_c \times \sqrt{\epsilon_r}) = 1.875 \times 10^7 / (f_c \times \sqrt{\epsilon_r})$ [m]. When the frequency of the spurious emission is equal to or greater than five times the cutoff frequency f_c , $X \leq c / (5 \times 4 \times f_c \times \sqrt{\epsilon_r}) = 1.5 \times 10^7 / (f_c \times \sqrt{\epsilon_r})$ [m] is more preferable. For filters that allow high-frequency signals of 6 GHz or lower used in the 5G communication system to pass therethrough, $X \leq 2.68 \times 10^{-3} / \sqrt{\epsilon_r}$ [m] is preferable, and $X \leq 2.14 \times 10^{-3} / \sqrt{\epsilon_r}$ [m] is more preferable.

As in the first embodiment, the closed pattern may be one conductor pattern **M1** formed of the conductor layer **12f** provided between the adjacent dielectric layers **11e** and **11f**.

As in the first variation of the first embodiment, the closed pattern may include the conductor patterns **M1** and **M4**, and the via wirings connecting the end portions of the conductor pattern **M1** and the end portions of the conductor pattern **M2**. The conductor pattern **M4** (a fourth conductor pattern) is formed of the conductor layer **12d** (a fourth conductor layer) different from the conductor layer **12f**. The closed pattern may be formed of three or more different conductor patterns.

The conductor layer **12e** in which the conductor pattern **M2** is provided and the conductor layer **12e** in which the conductor pattern **M3** is provided may be different conductor layers or may be the same conductor layer. The conductor layer **12d** in which the conductor pattern **M4** is provided may be a conductor layer different from or the same as the conductor layer **12e** in which the conductor patterns **M2** and **M3** are provided.

Although the low-pass filter has been described as an example of the filter, the filter may be a band-pass filter.

SECOND EMBODIMENT

FIG. **12** is a circuit diagram of a diplexer in accordance with a second embodiment. As illustrated in FIG. **12**, a diplexer **20** includes filters **22** and **24**. The filter **22** is connected between a common terminal Ant and a terminal **T1**, and the filter **24** is connected between the common terminal Ant and a terminal **T2**. An antenna **28** is connected to the common terminal Ant. The filter **22** is, for example, a low-pass filter (LPF), allows a high-frequency signal in a low band to pass therethrough and suppresses signals with other frequencies. The filter **24** is, for example, a high-pass filter HPF, and allows a high-frequency signal in a high band higher than the low band in frequency to pass therethrough, and suppresses signals with other frequencies.

The filter **22** may be the filter in accordance with any one of the first embodiment and the variation thereof. Although the example of the diplexer is described as an example of the multiplexer, the multiplexer may be a triplexer or a quadplexer.

Although the embodiment of the present invention has been described in detail above, the present invention is not limited to the specific embodiment, and various modifications and changes can be made within the scope of the gist of the present invention described in the claims.

What is claimed is:

1. A filter comprising:

a multilayer body in which a plurality of dielectric layers and a plurality of conductor layers are alternately stacked in a stacking direction;
an input terminal provided on a surface of the multilayer body;

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an output terminal provided on a surface of the multilayer body;

a first conductor pattern formed of a first conductor layer of the plurality of conductor layers, the first conductor pattern having a planar shape that forms at least a part of a closed pattern having an opening when viewed from the stacking direction;

a second conductor pattern formed of a second conductor layer different from the first conductor layer of the plurality of conductor layers, and connected to the input terminal, at least a part of the second conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, an end portion of the second conductor pattern being located within the opening of the closed pattern when viewed from the stacking direction, the second conductor pattern extending from an inside of the opening to an outside of the closed pattern when viewed from the stacking direction; and

a third conductor pattern formed of a third conductor layer that is same as or different from the second conductor layer and that is different from the first conductor layer of the plurality of conductor layers, and connected to the output terminal, at least a part of the third conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, the third conductor pattern not overlapping with the second conductor pattern when viewed from the stacking direction, an end portion of the third conductor pattern being located within the opening of the closed pattern when viewed from the stacking direction, the third conductor pattern extending from the inside of the opening to the outside of the closed pattern when viewed from the stacking direction,

wherein a first width, along the opening, of a region where the second conductor pattern overlaps the first conductor pattern when viewed from the stacking direction is smaller than a width, in a width direction of the first width, of the opening, and

wherein a second width, along the opening, of a region where the third conductor pattern overlaps the first conductor pattern when viewed from the stacking direction is smaller than a width, in a width direction of the second width, of the opening.

2. The filter according to claim 1, wherein the closed pattern is a pattern in which a path through which a current flows is closed.

3. The filter according to claim 1, wherein the second conductor pattern and the third conductor pattern are provided so as to face each other across the opening when viewed from the stacking direction.

4. The filter according to claim 1, wherein a planar shape of a region where the first conductor pattern and the second conductor pattern overlap each other is a substantially parallelogram when viewed from the stacking direction, and a planar shape of a region where the first conductor pattern and the third conductor pattern overlap each other is a substantially parallelogram when viewed from the stacking direction.

5. The filter according to claim 1, wherein the first conductor pattern and the second conductor pattern form a first capacitor, and wherein the first conductor pattern and the third conductor pattern form a second capacitor.

6. The filter according to claim 5, further comprising: a first inductor connected in parallel to the first capacitor between the input terminal and the output terminal; and

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- a second inductor connected in parallel to the second capacitor between the input terminal and the output terminal.
7. The filter according to claim 6,
 wherein a first parallel resonant circuit including the first capacitor and the first inductor forms an attenuation pole at a frequency higher than a passband, and
 wherein a second parallel resonant circuit including the second capacitor and the second inductor forms an attenuation pole at a frequency higher than the passband.
8. The filter according to claim 5, further comprising:
 a ground terminal provided on a surface of the multilayer body; and
 a third capacitor having a first end connected to the first conductor pattern and a second end connected to the ground terminal.
9. A multiplexer comprising:
 the filter according to claim 1.
10. The filter according to claim 1, wherein $X \leq 1.875 \times 10^7 / (f_c \times \sqrt{\epsilon_r})$ is satisfied, where X (m) is a maximum width of the closed pattern, f_c (Hz) is a high frequency end of a passband, and ϵ_r is a relative permittivity of each of the plurality of dielectric layers.
11. The filter according to claim 1, wherein the filter is a low-pass filter or a band-pass filter.
12. A filter comprising:
 a multilayer body in which a plurality of dielectric layers and a plurality of conductor layers are alternately stacked in a stacking direction;
 an input terminal provided on a surface of the multilayer body;
 an output terminal provided on a surface of the multilayer body;
 a first conductor pattern formed of a first conductor layer of the plurality of conductor layers, the first conductor pattern having a planar shape that forms a notch pattern surrounding a recess when viewed from the stacking direction;

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- a second conductor pattern formed of a second conductor layer different from the first conductor layer of the plurality of conductor layers, and connected to the input terminal, at least a part of the second conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, an end portion of the second conductor pattern being located within the recess when viewed from the stacking direction, the second conductor pattern extending from an inside of the recess to an outside of the notch pattern when viewed from the stacking direction;
- a third conductor pattern formed of a third conductor layer that is same as or different from the second conductor layer and that is different from the first conductor layer of the plurality of conductor layers, and connected to the output terminal, at least a part of the third conductor pattern overlapping with at least a part of the first conductor pattern when viewed from the stacking direction, the third conductor pattern not overlapping with the second conductor pattern when viewed from the stacking direction, an end portion of the third conductor pattern being located within the recess when viewed from the stacking direction, the third conductor pattern extending from the inside of the recess to the outside of the notch pattern when viewed from the stacking direction;
- a fourth conductor pattern formed of a fourth conductor layer different from the first conductor layer of the plurality of conductor layers; and
 a via wiring that penetrates through one or more dielectric layers of the plurality of dielectric layers and connects the first conductor pattern and the fourth conductor pattern,
 wherein the first conductor pattern, the fourth conductor pattern, and the via wiring form a closed pattern which surrounds an opening when viewed from the stacking direction.

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